

SEG2105 – Introduction to Software Engineering

AI-Assisted Software Development Project

Project Overview

The objective of this project is to design, implement, and evaluate a software application that uses modern AI code-generation tools. Students will select a project topic that interests them, such as an online banking system, an online store, a personal finance manager, a course registration system, a fitness tracker, a task management application, or another software system of similar complexity.

The project will provide students with hands-on experience in requirements engineering, software design, implementation, testing, and software quality assurance while exploring the strengths and limitations of AI-assisted software development.

Learning Objectives

By completing this project, students will be able to:

- Translate user requirements into effective AI prompts.
- Use AI code generation tools to create software artifacts.
- Evaluate and improve the quality of AI-generated code.
- Identify missing requirements and features in generated software.
- Modify and extend AI-generated code to satisfy project requirements.
- Test and validate software against its specified requirements.
- Document the software development process and the role of AI in development.

Project Requirements

1. Project Selection

Students must choose a software application of personal interest. Examples include:

- Online Banking System
- Online Store / E-Commerce Platform
- Event Management System
- Library Management System
- Student Course Registration System
- Personal Budget Tracker
- Fitness Tracking Application
- Restaurant Reservation System
- Movie Recommendation System

The selected project must contain sufficient functionality to demonstrate meaningful software engineering effort.

2. Requirements Definition

Students must create an initial requirements specification describing:

- System purpose
- Target users
- Functional requirements
- Non-functional requirements
- Assumptions and constraints

These requirements will serve as the basis for generating software using AI tools.

3. AI-Assisted Development

Students must use an AI code generation tool (e.g., ChatGPT, GitHub Copilot, Claude, Gemini, Cursor, Windsurf, or similar) to generate most of the source code.

Students are expected to:

- Experiment with multiple prompts.
- Refine prompts iteratively to improve software quality.
- Compare the results obtained from different prompt formulations.
- Document prompt evolution throughout the project.

The project report must include representative prompts and a discussion of their effectiveness.

4. Feature Enhancement

AI-generated code is unlikely to fully satisfy all requirements.

Students must:

- Identify missing, incorrect, or incomplete functionality.
- Modify and extend the generated code.
- Implement at least one significant feature that was not correctly produced by the AI tool.
- Justify the changes made to the generated code.

5. Testing and Validation

Students must verify that the final software satisfies the stated requirements.

At a minimum, students must:

- Develop and execute test cases.
- Document testing results.
- Demonstrate that major requirements have been implemented successfully.
- Fix critical defects discovered during testing.

6. Final Deliverables

Students must submit:

1. Source code repository.
2. Requirements specification.
3. Project report.
4. Test plan and testing results.
5. Prompt engineering log documenting interactions with the AI tool.
6. Demonstration video (5–10 minutes) showing the system in operation.

Project Report Contents

The report should include:

1. Introduction and project description.
2. Requirements specification.
3. AI tools used.
4. Prompt engineering process.
5. Analysis of generated code quality.
6. Description of manual enhancements and added features.
7. Testing and validation results.
8. Lessons learned regarding AI-assisted software engineering.
9. Conclusion.

Evaluation Criteria

Criterion	Weight
Requirements Specification	15%
Effective Use of AI and Prompt Engineering	20%
Software Functionality	25%
Manual Enhancements and Missing Feature Implementation	15%
Testing and Validation	15%
Report and Documentation	10%

Academic Integrity

Students may use AI tools extensively for code generation; however, all AI assistance must be documented. Students remain responsible for understanding, validating, testing, and modifying the

generated code. Submissions should clearly identify the AI tools used and provide evidence of the prompt engineering process.

Deadline

Project Submission Deadline: July 17, 2026, at 8 PM.

Late submissions will be subject to the course's late submission policy.